

IGCSE Chemistry 知识点

Summary of Chapters 12–16, 18–19

Chapter 12: Preparation of Salts 盐的制备

Core Concepts 核心概念

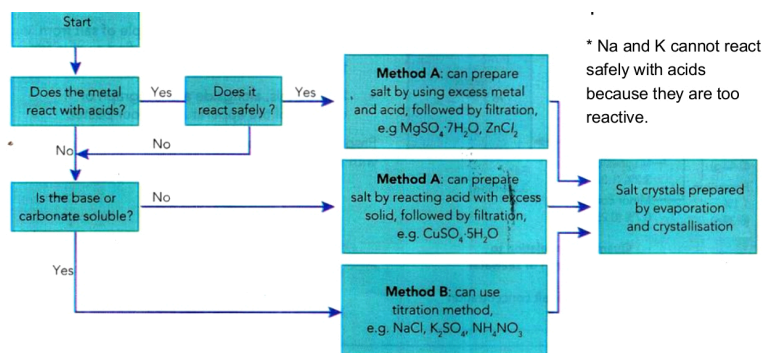
1. Solubility Rules 溶解性规则

- **All sodium, potassium, ammonium salts and nitrates are soluble.**
所有钠盐、钾盐、铵盐和硝酸盐都可溶。
- **Most chlorides are soluble, except silver chloride (AgCl) and lead(II) chloride (PbCl₂).**
大多数氯化物可溶，但氯化银和氯化铅除外。
- **Most sulfates are soluble, except barium sulfate (BaSO₄), lead(II) sulfate (PbSO₄), and calcium sulfate (CaSO₄).**
大多数硫酸盐可溶，但硫酸钡、硫酸铅和硫酸钙除外。
- **Most carbonates are insoluble, except sodium carbonate (Na₂CO₃), potassium carbonate (K₂CO₃), and ammonium carbonate ((NH₄)₂CO₃).**
大多数碳酸盐不溶，但碳酸钠、碳酸钾和碳酸铵除外。

2. Preparation of Soluble Salts 可溶盐的制备

Method A: Acid + excess metal / insoluble base / insoluble carbonate

方法A：酸 + 过量金属/不溶碱/不溶碳酸盐



- **Steps:** Warm acid → add excess solid → filter to remove excess solid → evaporate filtrate → crystallise → dry.
步骤：加热酸 → 加入过量固体 → 过滤除去过量固体 → 蒸发滤液 → 结晶 → 干燥。

- **Examples:**

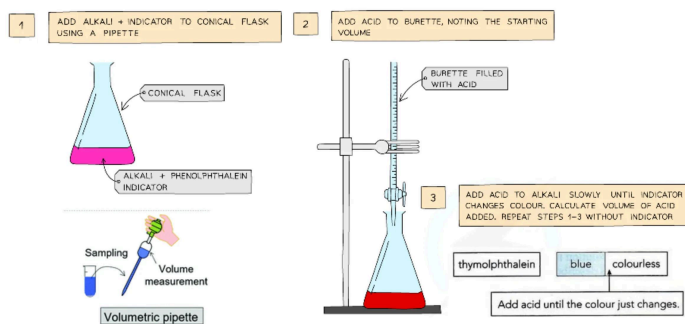
- Magnesium + sulfuric acid: $\text{Mg(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + \text{H}_2(\text{g})$
- Copper(II) oxide + sulfuric acid: $\text{CuO(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O(l)}$
- Zinc carbonate + sulfuric acid: $\text{ZnCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{H}_2\text{O(l)} + \text{CO}_2(\text{g})$

- **Key point:** Excess solid ensures all acid is consumed; filtration removes unreacted solid.
关键点：过量固体确保酸完全反应；过滤除去未反应的固体。

Method B: Titration (acid + alkali / soluble carbonate)

方法B：滴定法（酸 + 碱/可溶碳酸盐）

- Used when both reactants are soluble (e.g., $\text{NaOH} + \text{HCl}$).
用于两种反应物均可溶的情况。
- Steps: Pipette alkali + indicator → add acid from burette until colour change → repeat without indicator → evaporate → crystallise.
步骤：用移液管取碱液+指示剂 → 从滴定管加酸至颜色变化 → 不加指示剂重复 → 蒸发 → 结晶。



- **Example:** $\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$

- **Apparatus:** burette, pipette, conical flask.

仪器：滴定管、移液管、锥形瓶。

3. Preparation of Insoluble Salts 不溶盐的制备

- **Method: Precipitation** 方法：沉淀法

Steps: Mix two soluble salt solutions → filter → wash residue with distilled water → dry in oven.

步骤：混合两种可溶盐溶液 → 过滤 → 用蒸馏水洗涤沉淀 → 烘干。

- **Example:** $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$

Ionic equation: $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$

(White precipitate 白色沉淀)

- **Example:** $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$
 Ionic equation: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
 (White precipitate, turns grey in light 白色沉淀, 见光变灰)

4. Hydrated and Anhydrous Salts 水合盐与无水盐

- **Hydrated salt:** Contains water of crystallisation, e.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue), $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (pink), $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ (green).
 水合盐含有结晶水。
- **Anhydrous salt:** No water of crystallisation, e.g., CuSO_4 (white).
 无水盐不含结晶水。
- **Heating effect:** Hydrated salt $\rightarrow$ heat $\rightarrow$ anhydrous salt + water.
 加热水合盐得到无水盐和水。
- **Caution:** Do not heat too strongly when drying, otherwise the salt may decompose or lose water of crystallisation.
 干燥时避免过度加热, 否则盐可能分解或失去结晶水。

Key Difficulties 难点

- Memorising solubility rules and applying them to choose the correct method (excess solid vs. titration).
 记忆溶解性规则并选择正确方法 (过量固体法 vs. 滴定法)。
- Understanding why excess solid is used and how to separate the product.
 理解为何使用过量固体以及如何分离产物。
- Recognising hydrated salts and the effect of heating.
 识别水合盐及加热的影响。

Chapter 13: The Periodic Table 元素周期表

Core Concepts 核心概念

1. Arrangement of the Periodic Table 周期表的排列

- Elements are arranged in order of increasing **atomic (proton) number**.
 元素按原子序数 (质子数) 递增排列。
- **Groups:** Vertical columns; elements in the same group have the same number of outer electrons and similar chemical properties.
 族: 纵列; 同族元素具有相同的最外层电子数和相似的化学性质。

- **Periods:** Horizontal rows; elements in the same period have the same number of electron shells.
周期：横行；同周期元素具有相同的电子层数。

- **Main-group elements:** Groups I–VIII (excluding transition metals).
主族元素：第I–VIII族（不含过渡金属）。

2. Group I – Alkali Metals 第I族 – 碱金属

- **Physical properties:** Soft, low density, low melting points, silvery appearance.
物理性质：软、低密度、低熔点、银白色。
- **Trends down the group:**
 - Melting point decreases. 熔点降低。
 - Density generally increases (except K). 密度通常增大。
 - Reactivity increases. 反应性增强。
- **Reaction with water:** $2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$ ($M = Li, Na, K$)
 - Lithium: fizzes slowly, no flame. 锂：缓慢冒泡，无火焰。
 - Sodium: melts into a ball, moves rapidly, no flame. 钠：熔成小球，快速移动，无火焰。
 - Potassium: ignites hydrogen with a lilac flame. 钾：点燃氢气，产生淡紫色火焰。
- **Storage:** Stored under oil to prevent reaction with air and moisture.
储存：浸在油中以防与空气和水分反应。

3. Group VII – Halogens 第VII族 – 卤素

- **Physical properties:** Diatomic molecules (F_2, Cl_2, Br_2, I_2); colour intensity increases down the group; state changes from gas (Cl_2) to liquid (Br_2) to solid (I_2) at room temperature.
物理性质：双原子分子；颜色加深；室温下状态从气体到液体到固体。
- **Trends down the group:**
 - Melting and boiling points increase. 熔沸点升高。
 - Reactivity decreases. 反应性减弱。
- **Displacement reactions:** A more reactive halogen displaces a less reactive one from its halide solution.
置换反应：更活泼的卤素置换出较不活泼的卤素。

- $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$ (colourless $\rightarrow$ orange)
- $\text{Cl}_2 + 2\text{KI} \rightarrow 2\text{KCl} + \text{I}_2$ (colourless $\rightarrow$ yellow-brown)
- $\text{Br}_2 + 2\text{KI} \rightarrow 2\text{KBr} + \text{I}_2$
- **Test for chlorine:** Damp blue litmus paper turns red then bleached white.
氯气检验：湿润的蓝色石蕊试纸先变红后漂白变白。

4. Group VIII – Noble Gases 第VIII族 – 稀有气体

- **Properties:** Colourless, monatomic, unreactive (full outer shell: 8 electrons, except He with 2).
性质：无色、单原子、不活泼（最外层满电子）。
- **Uses:**
 - Helium: balloons, airships (light, unreactive).
 - Argon: filling light bulbs (inert at high temperature).
 - Neon: advertising signs (glows when electrified).

5. Transition Metals 过渡金属

- **General properties:** Hard, strong, high density, high melting points, less reactive than Group I/II.
一般性质：硬、强、高密度、高熔点、反应性低于I/II族。
- **Distinctive properties:**
 - Form coloured compounds (e.g., CuSO_4 blue, FeSO_4 green).
 - Act as catalysts (e.g., Fe in Haber process).
 - Have variable oxidation states (e.g., Fe^{2+} and Fe^{3+} ; CrO , Cr_2O_3 , CrO_3).
 - Form more than one oxide.

6. Trends Across a Period 周期趋势

- **Metallic character decreases** from left to right. 金属性从左到右减弱。
- **Non-metallic character increases.** 非金属性增强。
- **Bonding type changes:** metallic $\rightarrow$ giant covalent $\rightarrow$ simple molecular.
键型变化：金属键 $\rightarrow$ 巨型共价 $\rightarrow$ 简单分子。

- **Example – Period 3:** Na, Mg, Al (metallic); Si (giant covalent); P, S, Cl (simple molecular); Ar (monatomic).

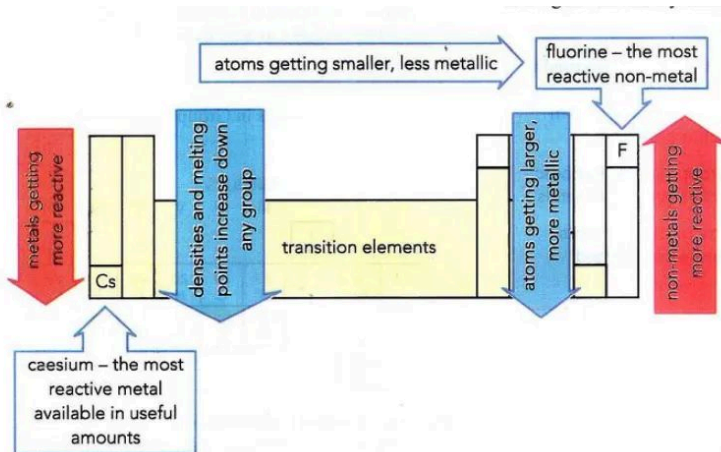


Figure 13.6: General trends in the Periodic Table, leaving aside the noble gases in Group VIII.

Key Difficulties 难点

- Relating group number to ionic charge (Group I $\rightarrow +1$, Group VII $\rightarrow -1$, etc.).
将族号与离子电荷关联。
- Understanding the trend in reactivity down Group I (increases) vs. Group VII (decreases).
理解I族反应性增强而VII族减弱的趋势。
- Remembering the distinctive properties of transition metals (coloured compounds, variable oxidation states, catalysis).
记住过渡金属的特性（有色化合物、可变氧化态、催化作用）。

Chapter 14: Metallic Elements and Alloys 金属元素与合金

Core Concepts 核心概念

1. Physical Properties of Metals 金属的物理性质

- **Electrical conductivity:** Due to delocalised (free) electrons that can move through the structure.
导电性：由于离域电子可在结构中自由移动。
- **Thermal conductivity:** Vibrations of metal atoms are passed through the structure.
导热性：金属原子的振动传递。
- **Malleability & Ductility:** Layers of metal atoms can slide over each other without breaking bonds.
可塑性和延展性：金属原子层可相互滑动而不断键。

- **Sonorous:** Ringing sound when struck (dense, closely packed atoms transmit vibrations).
有响声：敲击时发出响亮声音。
- **High melting/boiling points:** Strong metallic bonds require large energy to break.
高熔沸点：强金属键需要大量能量才能破坏。
- **Hardness and density:** Generally hard and dense (except Na, K which are soft).
硬度和密度：通常硬而密。

2. Reactions of Metals 金属的反应

- **With oxygen/air:**
 - $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (bright white flame)
 - $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ (protective oxide layer)
 - $3\text{Fe} + 2\text{O}_2 \rightarrow \text{Fe}_3\text{O}_4$ (in air) or $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
- **With water:**
 - Reactive metals (K, Na, Ca): $2\text{M} + 2\text{H}_2\text{O} \rightarrow 2\text{MOH} + \text{H}_2$ (cold water)
 - Mg: reacts very slowly with cold water, vigorously with steam: $\text{Mg} + \text{H}_2\text{O}(\text{g}) \rightarrow \text{MgO} + \text{H}_2$
 - Zn, Fe: react with steam: $\text{Zn} + \text{H}_2\text{O}(\text{g}) \rightarrow \text{ZnO} + \text{H}_2$; $3\text{Fe} + 4\text{H}_2\text{O}(\text{g}) \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
- **With dilute acid:**
 - $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$
 - $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
 - $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$
 - Cu, Ag, Au: no reaction.

3. Alloys 合金

- **Definition:** Mixtures of metals (or metals with non-metals) designed for specific properties.
定义：金属（或金属与非金属）的混合物，具有特定性能。
- **Structure:** Atoms of different sizes distort the regular lattice, making it harder for layers to slide → stronger and harder than pure metals.
结构：不同大小的原子扭曲晶格，使层滑动更困难 → 比纯金属更强更硬。
- **Examples:**

- **Brass:** Cu (70%) + Zn (30%) – harder than Cu, used for musical instruments, ornaments.
- **Bronze:** Cu (90%) + Sn (10%) – harder than Cu, used for statues, bells.
- **Mild steel:** Fe (99.7%) + C (0.3%) – stronger than pure Fe, used for car bodies.
- **Stainless steel:** Fe (74%) + Cr (18%) + Ni (8%) – hard, does not rust, used for cutlery, surgical instruments.
- **Solder:** Sn (50%) + Pb (50%) – low melting point, used for electrical connections.
- **Alloys still conduct electricity** because delocalised electrons are still free to move.
合金仍能导电，因为离域电子仍可自由移动。

Key Difficulties 难点

- Explaining why alloys are harder than pure metals in terms of atomic structure.
从原子结构解释合金为何比纯金属硬。
- Linking properties of metals to their uses (e.g., Al low density for aircraft, Cu conductivity for wiring).
将金属性质与用途关联。
- Remembering the composition and properties of common alloys.
记住常见合金的成分和性质。

Chapter 15: Reactivity of Metals 金属的反应性

Core Concepts 核心概念

1. The Reactivity Series 金属活动性顺序

- **Order (most reactive first):** K, Na, Ca, Mg, Al, (C), Zn, Fe, (H), Cu, Ag, Au
顺序（从高到低）：钾、钠、钙、镁、铝、（碳）、锌、铁、（氢）、铜、银、金
- **Reactivity** refers to the ease with which a metal forms its positive ion (loses electrons).
反应性指金属形成阳离子（失去电子）的难易程度。
- **Hydrogen** is included for comparison; metals above H can displace H from acids/water.
氢被纳入以便比较；氢以上的金属可从酸或水中置换出氢。

2. Reactions with Water and Acids 与水及酸的反应

- **Group I metals:** Violent reaction with cold water, producing hydroxide and H_2 .
- **Ca:** Moderate reaction with cold water.

- **Mg:** Slow with cold water, vigorous with steam.
- **Al:** Protective oxide layer slows reaction with acid; once removed, reacts.
- **Zn, Fe:** React with steam or dilute acids.
- **Cu, Ag, Au:** No reaction with water or dilute acids.

3. Displacement Reactions 置换反应

- **Definition:** A more reactive metal displaces a less reactive metal from a solution of its salt.
定义：更活泼的金属从其盐溶液中置换出较不活泼的金属。
- **Example:** $\text{Zn(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu(s)}$
Observations: blue solution fades, grey solid deposits.
现象：蓝色溶液变浅，灰色固体析出。
- **Example:** $\text{Cu(s)} + 2\text{AgNO}_3(\text{aq}) \rightarrow \text{Cu(NO}_3)_2(\text{aq}) + 2\text{Ag(s)}$
Observations: colourless solution turns blue, silver crystals form.
现象：无色溶液变蓝，银白色晶体析出。
- **Redox nature:** The more reactive metal loses electrons (oxidised), the less reactive metal ions gain electrons (reduced).
氧化还原本质：更活泼金属失去电子（被氧化），较不活泼金属离子得到电子（被还原）。
- **Ionic equation:** $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$

4. Aluminium's Apparent Low Reactivity 铝的表观低反应性

- Al reacts readily with oxygen to form a thin, adherent layer of Al_2O_3 .
铝易与氧反应形成致密的氧化铝薄膜。
- This layer protects the metal from further reaction with water or acid.
该层保护金属不与水或酸进一步反应。
- Once the layer is removed (e.g., by heating or scratching), Al reacts vigorously.
一旦去除氧化层（如加热或刮擦），铝会剧烈反应。

Key Difficulties 难点

- Memorising the reactivity series (use a mnemonic).
记忆活动性顺序（使用口诀）。
- Predicting displacement reactions and writing ionic equations.
预测置换反应并书写离子方程式。

- Understanding the role of the oxide layer in aluminium's behaviour.
理解氧化层在铝反应行为中的作用。
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Chapter 16: Extraction and Corrosion of Metals 金属的提取与腐蚀

Core Concepts 核心概念

1. Extraction Methods Based on Reactivity 基于反应性的提取方法

- **Metals below carbon in the series** (Zn, Fe, Cu): extracted by reduction with carbon (heating oxide with carbon/coke).
碳以下的金属：用碳还原（加热氧化物与碳/焦炭）。
- **Metals above carbon** (K, Na, Ca, Mg, Al): extracted by electrolysis of molten ore.
碳以上的金属：通过电解熔融矿石提取。
- **Very unreactive metals** (Ag, Au): found native (uncombined) in the Earth's crust.
极不活泼金属：以游离态存在于地壳中。

2. Extraction of Iron – Blast Furnace 铁的提取 – 高炉

- **Raw materials:** hematite (Fe_2O_3), coke (C), limestone (CaCO_3), air.
原料：赤铁矿、焦炭、石灰石、空气。
- **Key reactions:**
 - $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (heat)
 - $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
 - $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (reduction)
 - $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (decomposition)
 - $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ (slag formation – removes impurities)
- **Products:** Molten iron (about 95% pure) and slag (calcium silicate).
产物：熔融铁（约95%纯）和炉渣（硅酸钙）。

3. Extraction of Aluminium – Electrolysis 铝的提取 – 电解

- **Ore:** Bauxite (Al_2O_3).
矿石：铝土矿。
- **Process:**
 - Bauxite is purified to alumina (Al_2O_3).

- Alumina is dissolved in molten cryolite (Na_3AlF_6) to lower the melting point ($\sim 950^\circ\text{C}$).
- Electrolysis:
 - Cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al(l)}$ (reduction)
 - Anode: $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$ (oxidation); oxygen reacts with carbon anode to form CO_2 , so anodes need regular replacement.
- **Why expensive:** High electricity cost; anodes need frequent replacement.
为何昂贵：高电耗；阳极需频繁更换。

4. Corrosion of Iron – Rusting 铁的腐蚀 – 生锈

- **Rust:** Hydrated iron(III) oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.
铁锈：水合氧化铁。
- **Conditions required:** Oxygen (air) and water.
必要条件：氧气（空气）和水。
- **Accelerating factors:** Salt (e.g., seawater) and acid rain.
加速因素：盐和酸雨。

5. Rust Prevention 防锈方法

- **Barrier methods:** Painting, oiling/greasing, plastic coating, electroplating (with tin or chromium).
屏障法：喷漆、涂油、塑料涂层、电镀。
- **Galvanising:** Coating with zinc. Even if scratched, zinc corrodes preferentially (sacrificial protection) because Zn is more reactive than Fe.
镀锌：即使涂层破损，锌优先腐蚀（牺牲保护）。
- **Sacrificial protection:** Attaching blocks of a more reactive metal (e.g., Zn, Mg) to the iron structure; the more reactive metal corrodes instead of iron.
牺牲保护：连接更活泼金属块（如锌、镁），该金属腐蚀而铁被保护。

Key Difficulties 难点

- Explaining why aluminium must be extracted by electrolysis while iron can be reduced by carbon.
解释为何铝必须电解而铁可用碳还原。
- Writing balanced equations for the blast furnace reactions.
书写高炉反应的平衡方程式。

- Understanding the role of cryolite and why anodes burn away.
理解冰晶石的作用及阳极消耗的原因。
 - Distinguishing between barrier methods, galvanising, and sacrificial protection.
区分屏障法、镀锌和牺牲保护。
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Chapter 18: Introduction to Organic Chemistry 有机化学导论

Core Concepts 核心概念

1. Homologous Series 同系列

- **Definition:** A series of organic compounds with the same functional group, same general formula, similar chemical properties, and a trend in physical properties.
定义：具有相同官能团、相同通式、相似化学性质和物理性质递变的一系列有机化合物。
- **Examples:**
 - Alkanes: C_nH_{2n+2} (e.g., CH_4 , C_2H_6)
 - Alkenes: C_nH_{2n} (e.g., C_2H_4 , C_3H_6)
 - Alcohols: $C_nH_{2n+1}OH$ (e.g., CH_3OH , C_2H_5OH)
 - Carboxylic acids: $C_nH_{2n+1}COOH$ (e.g., $HCOOH$, CH_3COOH)

2. Functional Groups 官能团

- **Alkane:** C–C single bond (saturated)
- **Alkene:** C=C double bond (unsaturated)
- **Alcohol:** –OH (hydroxyl group)
- **Carboxylic acid:** –COOH (carboxyl group)
- **Ester:** –COO– (ester linkage)

3. Naming Organic Compounds 有机化合物命名

- **Prefix:** Number of carbon atoms (meth- 1, eth- 2, prop- 3, but- 4, pent- 5, hex- 6)
- **Suffix:** Indicates functional group (-ane, -ene, -anol, -anoic acid, -yl ... -anoate for esters)
- **Examples:**
 - $C_4H_{10} \rightarrow$ butane
 - $C_3H_6 \rightarrow$ propene

- $\text{C}_3\text{H}_7\text{OH} \rightarrow$ propanol
- $\text{C}_3\text{H}_7\text{COOH} \rightarrow$ butanoic acid

4. Structural Isomerism 结构异构

- **Definition:** Compounds with the same molecular formula but different structural formulae.
定义：分子式相同但结构式不同的化合物。
- **Example:** C_4H_{10} has two isomers: butane (straight chain) and methylpropane (branched).
例：丁烷和甲基丙烷。
- **Position isomerism:** Same functional group but different position (e.g., propan-1-ol vs. propan-2-ol).
位置异构：官能团位置不同。

5. Saturated vs. Unsaturated 饱和与不饱和

- **Saturated:** Contains only single bonds (alkanes).
饱和：只含单键。
- **Unsaturated:** Contains at least one double or triple bond (alkenes).
不饱和：含至少一个双键或三键。
- **Test:** Add bromine water; unsaturated compounds decolourise it (orange $\rightarrow$ colourless).
检验：加入溴水，不饱和化合物使其褪色。

Key Difficulties 难点

- Learning the naming rules and applying suffixes correctly.
学习命名规则并正确使用后缀。
- Drawing structural isomers and recognising position isomerism.
画出结构异构体并识别位置异构。
- Understanding the relationship between functional group and chemical properties.
理解官能团与化学性质的关系。

Chapter 19: Reactions of Organic Compounds 有机化合物的反应

(Based on the previously uploaded PPT “19 Reactions of Organic Compounds.pptx” and standard IGCSE syllabus)

(基于之前上传的PPT和标准IGCSE大纲)

Core Concepts 核心概念

1. Reactions of Alkanes – Substitution 烷烃的反应 – 取代

- **Combustion:**
 - Complete: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 - Incomplete: $2\text{CH}_4 + 3\text{O}_2 \rightarrow 2\text{CO} + 4\text{H}_2\text{O}$ (produces toxic CO)
- **Substitution with halogens:** Requires **ultraviolet (UV) light**.
与卤素取代：需要紫外线。
 - $\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{(UV)}} \text{CH}_3\text{Cl} + \text{HCl}$
 - Further substitution yields CH_2Cl_2 , CHCl_3 , CCl_4 .

2. Reactions of Alkenes – Addition 烯烃的反应 – 加成

- **Source:** Catalytic cracking of long-chain alkanes (e.g., $\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_8\text{H}_{18} + \text{C}_2\text{H}_4$).
来源：长链烷烃催化裂解。
- **Addition reactions (C=C opens):**
 - **With bromine:** $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$ (bromine water decolourised – test for unsaturation)
 - **With hydrogen (hydrogenation):** $\text{C}_2\text{H}_4 + \text{H}_2 \xrightarrow{\text{(Ni catalyst, 150-300}^\circ\text{C)}} \text{C}_2\text{H}_6$ (used to harden vegetable oils)
 - **With steam (hydration):** $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \xrightarrow{\text{(H}_3\text{PO}_4 \text{ catalyst, 300}^\circ\text{C, 6000 kPa)}} \text{C}_2\text{H}_5\text{OH}$ (industrial ethanol production)

3. Chemistry of Ethanol 乙醇的化学性质

- **Manufacture – two methods:**

Method	Fermentation	Hydration of ethene
Starting material	Glucose (renewable)	Ethene from petroleum (non-renewable)
Conditions	Yeast, anaerobic, $\sim 37^\circ\text{C}$	H_3PO_4 catalyst, 300°C , 6000 kPa
Equation	$\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$	$\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH}$

Advantages	Renewable, simple equipment	Fast, high purity
Disadvantages	Slow, needs distillation	Non-renewable, high pressure

- **Oxidation to carboxylic acid:**

- Reagent: acidified KMnO_4 (purple $\rightarrow$ colourless) or acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (orange $\rightarrow$ green).
- $\text{C}_2\text{H}_5\text{OH} + 2[\text{O}] \xrightarrow{(\text{KMnO}_4/\text{H}^+)} \text{CH}_3\text{COOH} + \text{H}_2\text{O}$

- **Combustion:** $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$

4. Carboxylic Acids and Esters 羧酸与酯

- **Carboxylic acids are weak acids:**

- With metal: $2\text{CH}_3\text{COOH} + \text{Mg} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2$
- With base: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$
- With carbonate: $2\text{CH}_3\text{COOH} + \text{CaCO}_3 \rightarrow (\text{CH}_3\text{COO})_2\text{Ca} + \text{H}_2\text{O} + \text{CO}_2$

- **Esterification:**

- $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \xrightleftharpoons{(\text{conc. H}_2\text{SO}_4, \text{ heat})} \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$
- Product: ethyl ethanoate (fruity smell).
- **Reversible reaction** ($\rightleftharpoons$); conc. H_2SO_4 acts as catalyst and dehydrating agent.

Key Difficulties 难点

- Distinguishing substitution (alkanes) from addition (alkenes).
区分取代（烷烃）和加成（烯烃）。
- Remembering specific conditions for each reaction (catalyst, temperature, pressure, UV light).
记住每个反应的具体条件。
- Comparing fermentation and hydration for ethanol production.
对比发酵法和水合法制乙醇。
- Understanding that esterification is reversible and the role of concentrated H_2SO_4 .
理解酯化反应的可逆性及浓硫酸的作用。

Overall Common Pitfalls & Exam Focus 总体易错点与考试重点

Common Pitfalls 常见易错点

1. **Confusing soluble and insoluble salts** – always refer to solubility rules before choosing a method.
混淆可溶与不溶盐 – 选择方法前先查溶解性规则。
2. **Mixing up group trends** – Group I reactivity increases down; Group VII decreases.
混淆族趋势 – I族反应性向下增强，VII族减弱。
3. **Forgetting the oxide layer on aluminium** – it explains why Al appears less reactive.
忘记铝的氧化层 – 这解释了铝为何表现不活泼。
4. **Misunderstanding alloy structure** – alloys are not mixtures of separate crystals but distorted lattices.
误解合金结构 – 合金不是分离晶体的混合物，而是扭曲的晶格。
5. **Writing incorrect ionic equations for displacement** – include state symbols and charges.
写错置换反应的离子方程式 – 需包含状态符号和电荷。
6. **Omitting conditions for organic reactions** – UV for substitution, Ni for hydrogenation, H_3PO_4 for hydration.
遗漏有机反应条件 – 取代需UV，氢化需Ni，水合需 H_3PO_4 。
7. **Confusing fermentation and hydration** – one is biological, the other industrial.
混淆发酵法和水合法 – 一个是生物法，一个是工业法。
8. **Forgetting that esterification is reversible** – use $\rightleftharpoons$ and mention conc. H_2SO_4 as dehydrating agent.
忘记酯化反应可逆 – 使用 $\rightleftharpoons$ 并说明浓硫酸的吸水作用。

Exam Focus 考试重点

- **Writing balanced symbol equations** for all reactions (including state symbols).
书写所有反应的平衡符号方程式（含状态符号）。
- **Stating reaction conditions** (catalyst, temperature, pressure, light).
说明反应条件。
- **Identifying reaction types** (substitution, addition, esterification, redox, precipitation).
识别反应类型。
- **Comparing methods** (e.g., ethanol production, salt preparation).
对比不同方法。

- **Describing tests** (bromine water for unsaturation, chlorine test with litmus, flame tests for Group I).
描述检验方法。
 - **Explaining trends** in the Periodic Table and reactivity series.
解释周期表和反应性顺序的趋势。
 - **Applying solubility rules** to choose the correct salt preparation method.
应用溶解性规则选择正确的盐制备方法。
 - **Understanding rust prevention** mechanisms (barrier, galvanising, sacrificial).
理解防锈机制。
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可能不全